NFL Big Data Bowl 2027

Complete project guide in simple English

Please read first: Kaggle's Rules and Timeline tabs could not be read by my tools (they load with JavaScript). Items marked **[Confirmed]** come from the Kaggle Overview/Data pages. Items marked **[Typical]** come from earlier Big Data Bowl years. Items marked **[My advice]** are recommendations. **Always check the Rules tab for final deadlines, tracks, and limits.**

1. Competition at a glance

Topic	Detail
Host	The National Football League (NFL) [Confirmed]
Type	Featured Hackathon [Confirmed]
Theme	Find non-obvious links between NFL Combine sensor tracking data and regular-season NFL game performance [Confirmed]
Prize pool	$100,000. No Kaggle points or medals [Confirmed]
Timing	Recently started, about 3 months to close when checked [Confirmed]
Link	https://www.kaggle.com/competitions/nfl-big-data-bowl-2027

2. Step-by-step requirements

Step 1: Understand the goal
Study how a player's Combine tracking and testing results relate to how he performs in real NFL games. Find useful, non-obvious patterns. [Confirmed]

Step 2: Join the competition
Sign in to Kaggle, open the competition page, click Join, and accept the rules. You must accept the rules before submitting. [Confirmed that rules acceptance is standard on Kaggle]

Step 3: Download the data
Use the Data tab (details in Section 3). [Confirmed]

Step 4: Choose a track
Past years had a University track (students only, proof may be needed) and an open track. Check the Rules tab for 2027. [Typical]

Step 5: Form a team
Past years allowed teams. All members must accept the rules. Check the team and merge deadlines. [Typical]

Step 6: Build a public Kaggle Notebook
Your analysis code with clear explanations. A private notebook attached to a submission is usually made public after the deadline. [Typical]

Step 7: Write your Kaggle Writeup
Title, subtitle, and a clear explanation of your work, usually under 2,000 words, with images and your notebook attached. [Typical]

Step 8: Submit before the deadline
Click Submit on the Writeup. Drafts are not judged. Deadlines are usually 11:59 PM UTC. [Typical]

Step 9: Know how you are judged
Judges are usually NFL analytics staff. Past scoring covered football value, data science quality, visuals, and communication. [Typical]

3. The dataset: where to get it and what is inside

Where to get it: from the competition's **Data tab** on Kaggle (after you join and accept the rules): https://www.kaggle.com/competitions/nfl-big-data-bowl-2027/data
Two ways to use it:

- **Inside Kaggle (easiest):** create a Notebook from the competition. The data is attached automatically under /kaggle/input/nfl-big-data-bowl-2027/ with no download needed.
- **On your own computer:** click Download All on the Data tab, or use the Kaggle CLI: *kaggle competitions download -c nfl-big-data-bowl-2027* (needs a Kaggle API token from your account settings).

What the data contains [Confirmed]: high-frequency optical tracking, standardized physical testing results, and multi-season NFL game performance data for **510 rookie prospects** who were drafted or signed between **2023 and 2025**. There are **9 CSV files**, about **2.2 GiB** in total, linking pre-draft physical traits to regular-season outcomes.

File	What it holds	Size
combine_results.csv	Official body measurements, standardized testing splits, and Next Gen Stats draft grades from the Combine. One row per player (510 rows, 18 columns). Columns include draft_year, combine_position, combine_weight.	41 KiB
combine_tracking.csv	10 Hz (10 readings per second) sensor tracking from 6,310 Combine drill attempts, such as 40-yard dashes and position-specific agility and route drills (463,189 rows, 14 columns). Keys: event_id, nfl_id, time. nfl_id is empty for ball frames.	79.9 MiB
games.csv	Game-level information. Key: game_id.	see Data tab
game_tracking_2023.csv, game_tracking_2024.csv, game_tracking_2025.csv	10 Hz in-game tracking frames for all rookie-cohort players in each NFL season. Linked by game_id and nfl_id.	largest files
Other files (3 of the 9)	A players file (referenced by nfl_id) and further game-performance files. I could not see their exact names. Check the Data tab for the full file list and column dictionary.	see Data tab

How the files connect: *nfl_id* links a player across Combine and game files. *game_id* links game tracking to games.csv. The full column dictionary is on the Data tab.
Optional outside data [Typical, confirm in the Rules tab that external public data is allowed]:

- nflverse (free NFL play-by-play and stats): https://github.com/nflverse
- Pro Football Reference (stats and draft history): https://www.pro-football-reference.com
- Older Big Data Bowl data homepage (format examples): https://github.com/nfl-football-ops/Big-Data-Bowl

4. Is this a Machine Learning problem or an AI problem?

Short answer: use Machine Learning (and statistics). This is a data science and analytics problem.

- **AI** is the big umbrella (chatbots, robots, language models). **Machine Learning** is the part of AI that learns patterns from data. That is exactly what this task needs.
- Your data is **tabular and time-series sensor data** about 510 players. That is a small dataset. Classic ML works best here. Large neural networks or LLMs will likely overfit and are not needed. [My advice]
- The task is open-ended: **find and explain links**. So **explainability** (why a result happens) matters as much as accuracy, because NFL analysts will judge how useful and believable your findings are.

Typical problem framing [My advice]:

- **Regression:** predict a game-performance number (for example, speed, snaps, or a performance score) from Combine features.
- **Classification:** predict a category (for example, starter versus backup, or a high versus low performer).
- **Clustering / similarity:** group players with similar movement profiles and see how groups perform in games.
- **Feature discovery:** turn raw 10 Hz sensor data into new, meaningful metrics (acceleration burst, top speed, change-of-direction quality) and test whether they beat standard Combine numbers.

5. Which model should you use?

Stage	Model	Why
Baseline	Linear / Ridge / Elastic Net regression, Logistic regression	Fast, simple, easy to explain. Gives a score to beat.
Main model	LightGBM, XGBoost, or CatBoost (gradient boosting)	Best all-round for tabular data, handles messy features, strong with small data.
Cross-check	Random Forest	Stable, hard to break, good second opinion.
Explain it	SHAP values, feature importance, partial dependence plots	Shows which Combine traits drive game performance. Judges value this.
Position effects	Mixed-effects models (statsmodels) with position as a group	Players at different positions behave differently. This avoids misleading overall results.
Pattern finding	PCA, k-means or hierarchical clustering	Finds player types from movement profiles.
Optional, advanced	1D CNN or GRU on the raw 10 Hz sequences	Only if you have time. With 510 players, keep it small and compare against the boosting model.

Recommended path [My advice]: (1) simple baseline, (2) engineered features from the tracking data plus LightGBM, (3) SHAP to explain, (4) test whether your new sensor-based features beat standard Combine numbers (40-yard time, weight, etc.), and (5) tell a clear football story with charts.

Important cautions:

- **Small data:** use cross-validation grouped by player so the same player never appears in both training and test folds.
- **Data leakage:** do not use information from the future (for example, later seasons) to predict earlier results.

- **Position and playing-time bias:** compare players within the same position group and account for how many snaps each player played.

6. Tools and software required

Purpose	Tool	Notes
Platform	Kaggle account and Kaggle Notebooks	Free CPU/GPU, data attached automatically, required for submission.
Language	Python 3.10+	Standard for this work. R is also fine.
Data handling	pandas, NumPy, Polars (or DuckDB), PyArrow	Polars or DuckDB handle the large game_tracking files faster. Convert CSV to Parquet to save memory.
ML	scikit-learn, LightGBM, XGBoost, CatBoost	Core modeling libraries.
Explainability	SHAP	Explains model decisions.
Statistics	statsmodels, SciPy	Mixed-effects models and statistical tests.
Time-series features	tsfresh or tsfel (optional), SciPy signal	Auto-create features from 10 Hz sensor streams.
Visualization	Matplotlib, Seaborn, Plotly	Plotly for interactive field or movement charts.
Football plotting (optional)	mplsoccer-style pitch drawing code or custom Matplotlib field	For animating player movement.
Environment	Jupyter / VS Code, Git and GitHub	Local work and version control.
Kaggle CLI	kaggle (pip install kaggle)	Download data and manage notebooks from the terminal.
Writing and slides	Markdown (Kaggle Writeup editor), Google Docs, optional video tool	For the Writeup and media gallery.

7. Important links, APIs, and extras

Resource	Link / Info
Competition page	https://www.kaggle.com/competitions/nfl-big-data-bowl-2027
Data tab	https://www.kaggle.com/competitions/nfl-big-data-bowl-2027/data
Rules tab (read fully)	https://www.kaggle.com/competitions/nfl-big-data-bowl-2027/rules
Kaggle API (CLI)	https://www.kaggle.com/docs/api (create a token in Settings > API)
NFL Big Data Bowl page	https://operations.nfl.com/gameday/analytics/big-data-bowl
Past Big Data Bowl data format	https://github.com/nfl-football-ops/Big-Data-Bowl
nflverse (optional outside data)	https://github.com/nflverse (Python: nfl_data_py / nflreadpy)
SHAP documentation	https://shap.readthedocs.io

Resource	Link / Info
LightGBM documentation	https://lightgbm.readthedocs.io
Past winning notebooks and writeups	Search the Code and Writeups tabs of earlier Big Data Bowl competitions on Kaggle for structure and style ideas.

APIs: no paid API is required. The Kaggle API/CLI is optional for downloading data. If outside data is allowed, nflverse packages need no key.
Computer needs: The data is about 2.2 GiB. Work in Kaggle Notebooks if your laptop has under 16 GB of RAM, or load files one at a time and convert them to Parquet.

8. Suggested work plan [My advice]

- **Week 1:** Join, read rules, download data, explore each file, and understand the keys.
- **Week 2:** Clean data, build baseline, and engineer features from Combine tracking.
- **Weeks 3-5:** Link Combine features to game performance, train LightGBM, explain with SHAP, and test by position.
- **Weeks 6-8:** Polish visuals, find your key football insight, and write the Writeup.
- **Final days:** Clean the notebook, check word count, make the notebook public, and submit early (not on the last day).

9. Final checklist

■ Read the Rules tab fully and note deadlines
■ Accept the rules and finish team setup
■ Download or attach the 9 data files
■ Pick a clear question linking Combine data to game results
■ Build baseline, then main model, then explain it
■ Create clear charts a coach would understand
■ Write the Writeup within the word limit
■ Attach a public notebook and submit before the deadline

Prepared from the Kaggle Overview and Data pages as of Oct 2026. Confirm all rules and dates on Kaggle.